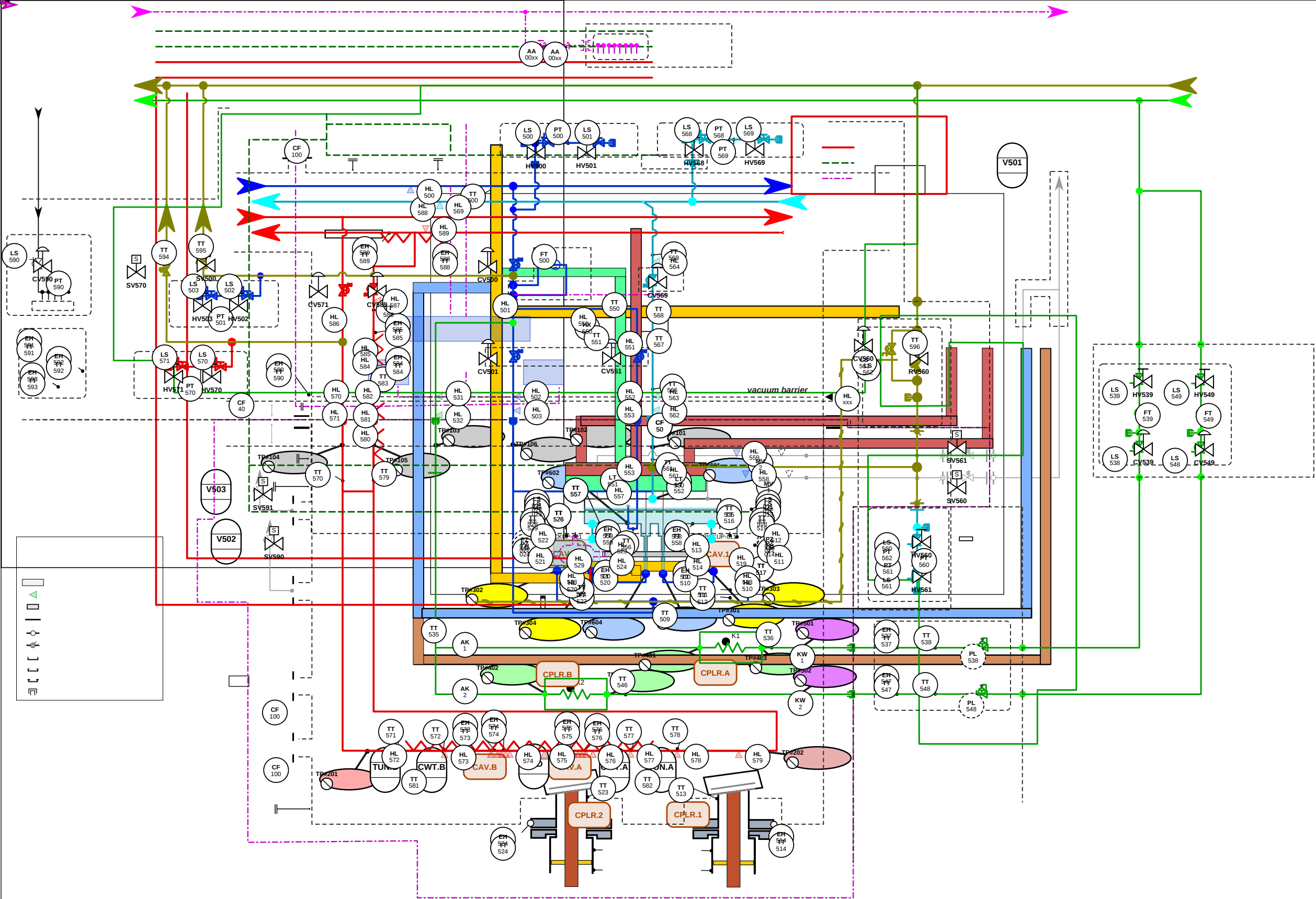


INSTRUMENT BUBBLES (field mounted)

- LB cryo instrument
- RFCELL instrument
- LBI-specific instrument
- Protection / safety (dashed)
- Tag = [variable][function]-[loop]
e.g. TT-514: T=temperature, T=transmitter

VALVES & EQUIPMENT

- Hand valve (HV)
- Control valve (CV)
- Solenoid valve (SV)
- Relief valve (RV)
- Vessel / tank
- Cavity / coupler body
- Heat exchanger
- Heat-load callout (HL)
- Terminal / scope point
- Vacuum barrier / scope boundary



PROCESS LINE CLASSES (colour = service)

- Class A - 4.5 K / 3 bar (LHe supply)
- Class B - 3.5 K / 27 mbar (2 K LP return)
- Class D - 40 K / 14 bar (Thermal shield)
- Class E - 60 K / 13 bar (60 K return / guard)
- Water - DI cooling water (Coupler / FREIA)
- QINFRA - Scope division (Infrastructure)
- Inst. air - 6 (5-7) bar(g) (Pneumatic signal)

PROJECT	MINERVA CryoCell - SCK CEN
TITLE	MINERVA CryoCell - QCELL / LB Cryogenic Flo...
DRAWING No.	SCK CEN/84836013
STANDARD	ISO 10628 / ISA-5.1 - A3 (420x297 mm)
REV / DATE	v2.0 - rebuilt 2025